Charge conservation fixes the expansion of Bohmian congruences

Mir Shafayeth Jamil
Jamal Nazrul Islam Center for Advanced Research,
University of Chittagong, Chittagong 4331, Bangladesh
22210012@std.cu.ac.bd

Abstract

The quantum Raychaudhuri equation obtained by replacing classical geodesics with de Broglie–Bohm trajectories has been used to argue that quantum effects prevent geodesic focusing and remove the big-bang singularity. We show that the normalized Bohmian phase flow, $\hat{u}_\mu = \nabla_\mu S/M$ with $M = \sqrt{-(\nabla S)^2}$, is the unit flow of the conserved Klein–Gordon current $j^\mu = \mathcal{R}^2\nabla^\mu S/\hbar$. Current conservation then fixes the expansion of the flow in closed form, $\theta = -d\ln(\mathcal{R}^2 M)/d\tau$: neighbouring trajectories spread at exactly the rate at which the conserved density is diluted along them. Evaluated on this congruence, the quantum Raychaudhuri equation is the once-differentiated continuity equation and supplies no separate evolution law for θ, so focusing must be decided by the Klein–Gordon and Einstein equations. For a minimally coupled complex scalar we find that the flow has zero vorticity, that unbounded defocusing can occur only at wave-function nodes or where the effective mass vanishes, that the quantum potential cannot reverse null focusing for any state, and that the timelike convergence is bounded below by the classical mass-term value $-8\pi G m^2\mathcal{R}^2/\hbar^2$. In spatially flat Friedmann–Robertson–Walker cosmology the conserved charge imposes a dust-like floor on the energy density, which makes the Hubble rate strictly monotonic, excludes any bounce or recollapse, and drives the scale factor to zero in finite proper time along the Bohmian congruence. The separate mechanism based on a non-minimal ξR coupling lies outside this analysis.

PACS: 04.20.Cv, 04.20.Dw, 98.80.Qc, 03.65.Ta
Keywords: Raychaudhuri equation; de Broglie–Bohm theory; quantum potential; singularity theorems; energy conditions

1 Introduction

The Raychaudhuri equation [1, 2] governs the rate of change of the expansion θ of a congruence, and with an energy condition converts convergence into geodesic incompleteness through the singularity theorems [3, 4, 5]. Das [6] asked whether quantum mechanics alters this at the level of the matter whose trajectories form the congruence, replacing classical geodesics with de Broglie–Bohm trajectories [7, 8, 9, 10]: with $\psi = \mathcal{R}e^{iS/\hbar}$, the velocity field $u_a = \partial_a S/m$ defines a first-order flow whose expansion equation, the quantum Raychaudhuri equation (QRE), acquires terms of order $\hbar^2$, and the no-crossing property of the flow was taken to prevent conjugate points and hence focusing. On a Friedmann–Robertson–Walker (FRW) background the construction was reported to remove the big-bang singularity and to generate a cosmological-constant-like term [11]. The two effects there come from distinct terms: the cosmological-constant term from the quantum potential $\Box\mathcal{R}/\mathcal{R}$, the singularity removal from a curvature term proportional to ξR, the non-minimal coupling of

the Klein–Gordon field, which on FRW produces an H^4 contribution that makes the age integral diverge. We keep the two separate and treat the minimally coupled field, $\xi = 0$.

That flow is not normalized: the quantum Hamilton–Jacobi (QHJ) equation gives $(\nabla S)^2 = -m^2 + \hbar^2 \Box\mathcal{R}/\mathcal{R} \neq -m^2$, so $g_{\mu\nu} + u_\mu u_\nu$ is not a projector, $\nabla_\mu u^\mu$ is not a volume-expansion rate, and the trajectories need not remain timelike [12]. The remedy, noted there, is to normalize the flow,

$$\hat{u}_\mu \;=\; \frac{\nabla_\mu S}{M}, \qquad M \;\equiv\; \sqrt{-(\nabla S)^2} \;=\; \sqrt{m^2 + 2m\mathcal{Q}}\,, \tag{1}$$

with $\mathcal{Q}$ the quantum potential. Wherever $M^2 > 0$ this is a unit timelike vector whose projector, expansion, shear and vorticity are the standard objects of congruence theory [13, 14, 15]. The normalized congruence and its Raychaudhuri equation appear in Ref. [12], but what that equation determines about focusing was left open. Related work quantizes the congruence itself [16], examines the Bohmian deviation equation on a fixed background [17], and computes the focusing corrections from quantized metric fluctuations [18, 19].

The vector $\mathcal{R}^2\nabla^\mu S$ is the conserved Klein–Gordon current of the complex field, so the normalized phase flow (1) is the unit flow of a conserved current, and charge conservation fixes its expansion. Along the flow,

$$\theta \;=\; -\,\frac{d}{d\tau}\ln\!\big(\mathcal{R}^2 M\big), \tag{2}$$

where $d/d\tau = \hat{u}^\mu\nabla_\mu$.

Here $\theta = \nabla_\mu\hat{u}^\mu$ is the fractional rate of growth of the volume of a bundle of Bohmian trajectories per unit proper time τ along them, $\mathcal{R}^2$ is the field intensity, and M, the magnitude of the phase gradient, is the rest mass corrected by the quantum potential; $\mathcal{R}^2 M/\hbar$ is the rest-frame density of the conserved Klein–Gordon charge. Equation (2) states that the trajectories spread at exactly the rate at which that charge is diluted along them. It is $\nabla_\mu j^\mu = 0$ for a flow whose velocity is the direction of the current itself, and it fixes θ from $\mathcal{R}$ and M with no free constant.

The consequence must be stated precisely. That $\theta = -d\ln n/d\tau$ for the flow of any conserved density, and that the Raychaudhuri equation is a geometric identity for any congruence, are standard; neither is claimed as new, nor is the continuity equation uninformative. The claim is that, for this construction, the QRE evaluated on the normalized phase flow is the once-differentiated continuity relation rather than an independent evolution equation for θ. Because the flow is accelerated whenever $\mathcal{Q} \neq 0$, the Raychaudhuri identity does not close into a focusing inequality as it does for geodesic matter obeying an energy condition, and Eq. (2) states what its indefinite terms amount to in aggregate. Whether the congruence focuses is then a question about the growth of $\mathcal{R}^2 M$, to be answered by the Klein–Gordon and Einstein equations. The unnormalized expansion $\Box S/m$ is fixed by the current in the same way.

Three results follow. Kinematically (Sec. 3), the normalized flow is a gradient flow, so its vorticity vanishes identically; since vorticity is the only term in the Raychaudhuri equation with a definite defocusing sign, the no-crossing property removes rather than supplies an obstruction to focusing. Its acceleration is $a_\mu = -h_\mu{}^\nu\nabla_\nu \ln M$, the relativistic quantum force, and by (2) unbounded defocusing requires $\mathcal{R}^2 M \to 0$, a node of the field or the vanishing of the effective mass. Dynamically (Sec. 4), with the field gravitating, the null contraction of its stress tensor is a sum of squares for every state, with the quantum potential entering only through a positive factor, and the timelike convergence obeys

$$R_{\mu\nu}\hat{u}^\mu\hat{u}^\nu \;\geq\; -\,\frac{8\pi G\, m^2\mathcal{R}^2}{\hbar^2}, \tag{3}$$

the classical mass-term value. Cosmologically (Sec. 5), in flat FRW the conserved charge $C = \mathcal{R}^2 M a^3$

gives the chain

$$\begin{aligned}\nabla_\mu j^\mu = 0 \;&\Longrightarrow\; \mathcal{R}^2 M a^3 = C \;\Longrightarrow\; \rho \geq \frac{2mC}{\hbar^2 a^3} \\ &\Longrightarrow\; H^2 \geq \frac{16\pi G\, mC}{3\hbar^2 a^3} \\ &\Longrightarrow\; a \to 0 \text{ in finite proper time,}\end{aligned} \tag{4}$$

for every $C \neq 0$, with the Hubble rate strictly monotonic and no bounce or recollapse, while for $C = 0$ the Bohmian congruence does not exist.

The field is a classical, minimally coupled complex scalar, the object underlying Refs. [6, 11, 12]. The identities of Secs. 3 and 4 hold in any spacetime on the domain where the phase gradient is timelike; the cosmological result requires FRW symmetry. The homogeneous equations of a charged scalar in FRW are known in hydrodynamic form from scalar-field dark-matter cosmology [20, 21]; new here is their reading as the Bohmian congruence and the consequence for the QRE. The ξR mechanism of Ref. [11] is absent at $\xi = 0$ and is not addressed.

Conventions: signature $(-,+,+,+)$, $c = 1$, $\hbar$ explicit, and $\mu \equiv m/\hbar$. The curvature conventions are

$$[\nabla_\mu, \nabla_\nu]V^\rho = R^\rho{}_{\sigma\mu\nu}V^\sigma, \qquad R_{\mu\nu} = R^\rho{}_{\mu\rho\nu},$$

with $G_{\mu\nu} = 8\pi G\, T_{\mu\nu}$. An overdot in Sec. 5 denotes $d/d\tau$ with τ cosmic time; elsewhere $d/d\tau \equiv \hat{u}^\mu \nabla_\mu$.

2 Field Equations and the Normalized Phase Flow

2.1 Polar variables

We take the minimally coupled complex scalar

$$\mathcal{S}_m = \int d^4x\sqrt{-g}\Big[- g^{\mu\nu}\nabla_\mu\phi^*\nabla_\nu\phi - \mu^2\phi^*\phi\Big], \tag{5}$$

the minimal field content admitting a nontrivial polar phase, with field equation $(\Box - \mu^2)\phi = 0$. Write $\phi = \mathcal{R}e^{iS/\hbar}$ with $\mathcal{R} \geq 0$ and S real [8, 9], so that

$$\nabla_\mu\phi = \Big(\nabla_\mu\mathcal{R} + \frac{i}{\hbar}\mathcal{R}\,\nabla_\mu S\Big)e^{iS/\hbar}, \tag{6}$$

and, applying ∇^μ once more,

$$\begin{aligned}\Box\phi &= \Big[\Box\mathcal{R} + \frac{i}{\hbar}\nabla^\mu\mathcal{R}\,\nabla_\mu S + \frac{i}{\hbar}\mathcal{R}\,\Box S\Big]e^{iS/\hbar} \\ &\quad + \Big(\nabla_\mu\mathcal{R} + \frac{i}{\hbar}\mathcal{R}\,\nabla_\mu S\Big)\frac{i}{\hbar}\nabla^\mu S\, e^{iS/\hbar} \\ &= \Big[\Box\mathcal{R} + \frac{2i}{\hbar}\nabla_\mu\mathcal{R}\,\nabla^\mu S + \frac{i}{\hbar}\mathcal{R}\,\Box S - \frac{\mathcal{R}}{\hbar^2}(\nabla S)^2\Big]e^{iS/\hbar}.\end{aligned} \tag{7}$$

Inserting (7) into $(\Box - \mu^2)\phi = 0$ and dividing by $e^{iS/\hbar}$, the real and imaginary parts vanish separately:

$$\begin{aligned}&\text{Re}: \quad \Box\mathcal{R} - \frac{\mathcal{R}}{\hbar^2}(\nabla S)^2 - \mu^2\mathcal{R} = 0, \\ &\text{Im}: \quad \frac{2}{\hbar}\nabla_\mu\mathcal{R}\,\nabla^\mu S + \frac{1}{\hbar}\mathcal{R}\,\Box S = 0.\end{aligned} \tag{8}$$

The imaginary part multiplied by $\hbar\mathcal{R}$ is a total divergence,

$$\nabla_\mu\big(\mathcal{R}^2\nabla^\mu S\big) = 2\mathcal{R}\,\nabla_\mu\mathcal{R}\,\nabla^\mu S + \mathcal{R}^2\Box S = 0. \tag{9}$$

and the real part, with $\mu^2 = m^2/\hbar^2$, gives

$$(\nabla S)^2 = -m^2 + \hbar^2\,\frac{\Box\mathcal{R}}{\mathcal{R}} \;\equiv\; -M^2, \tag{10}$$

the QHJ equation, the second equality defining the effective mass M. With the covariant quantum potential [10, 22]

$$\mathcal{Q} \;\equiv\; -\frac{\hbar^2}{2m}\frac{\Box\mathcal{R}}{\mathcal{R}}, \qquad \text{so that} \qquad M^2 = m^2 + 2m\mathcal{Q}, \tag{11}$$

the phase gradient has squared norm $-M^2$, reducing to the classical mass shell at $\mathcal{Q} = 0$. Equation (9) is the conservation of the Klein–Gordon current $j^\mu = \mathcal{R}^2\nabla^\mu S/\hbar$, the Noether current of the global phase symmetry of (5). Equations (9) and (10) are the entire dynamical content of the matter sector.

2.2 The normalized flow and its domain

Normalizing the phase flow,

$$\hat{u}_\mu \;\equiv\; \frac{\nabla_\mu S}{M}, \qquad \hat{u}_\mu\hat{u}^\mu = \frac{(\nabla S)^2}{M^2} = -1, \tag{12}$$

defines a unit vector on the hydrodynamic domain

$$\mathcal{D} \;=\; \{\,x:\; M^2(x) > 0\,\} \;=\; \{\,x:\; \mathcal{Q}(x) > -m/2\,\}, \tag{13}$$

by (11). There $\nabla_\mu S$ is timelike and, on the positive-frequency branch, $\hat{u}^\mu$ is future directed, and $h_{\mu\nu} = g_{\mu\nu} + \hat{u}_\mu\hat{u}_\nu$ is a projector orthogonal to the flow,

$$\begin{aligned} h_{\mu\nu}\hat{u}^\nu &= \hat{u}_\mu + \hat{u}_\mu\big(\hat{u}_\nu\hat{u}^\nu\big) = \hat{u}_\mu - \hat{u}_\mu = 0, \\ h_\mu{}^\alpha h_\alpha{}^\nu &= \big(\delta_\mu{}^\alpha + \hat{u}_\mu\hat{u}^\alpha\big)\big(\delta_\alpha{}^\nu + \hat{u}_\alpha\hat{u}^\nu\big) \\ &= \delta_\mu{}^\nu + 2\hat{u}_\mu\hat{u}^\nu + \hat{u}_\mu\big(\hat{u}^\alpha\hat{u}_\alpha\big)\hat{u}^\nu = h_\mu{}^\nu, \end{aligned} \tag{14}$$

so that $\theta = \nabla_\mu\hat{u}^\mu$, the shear and the vorticity are the standard objects [13, 14, 15] and every kinematic identity of congruence theory holds. Outside $\mathcal{D}$ no timelike congruence can be assigned to the phase flow in any normalization: the failure of the unnormalized field becomes a domain condition on the normalized one. The boundary $\partial\mathcal{D}$, where $\mathcal{Q} \to -m/2$, reappears below as the locus of both the divergences of the expansion and the limit of timelike defocusing.

3 Kinematical Identity and Charge Conservation

3.1 The expansion from charge conservation

On $\mathcal{D}$ the current is timelike, and with $\nabla^\mu S = M\hat{u}^\mu$,

$$j^\mu = \frac{\mathcal{R}^2\nabla^\mu S}{\hbar} = n\,\hat{u}^\mu, \qquad n \equiv \frac{\mathcal{R}^2 M}{\hbar} \;>\; 0, \tag{15}$$

so $j^\mu j_\mu = -n^2$ and n is the rest-frame density of the current; in the normalization of (5) the rest-frame particle number density is $2n/\hbar$. Expanding the divergence of (15),

$$\nabla_\mu j^\mu = \nabla_\mu\big(n\hat{u}^\mu\big) = \hat{u}^\mu\nabla_\mu n + n\,\nabla_\mu\hat{u}^\mu = \dot{n} + n\theta, \tag{16}$$

with $\dot{n} \equiv dn/d\tau$. The left-hand side vanishes by (9) and $n > 0$ on $\mathcal{D}$, so

$$\theta = -\frac{\dot{n}}{n} = -\frac{d}{d\tau}\ln n = -\frac{d}{d\tau}\ln\Big(\frac{\mathcal{R}^2 M}{\hbar}\Big). \tag{17}$$

The constant $\hbar$ drops out, and with $M = \sqrt{m^2 + 2m\mathcal{Q}}$,

$$\theta \;=\; -\frac{d}{d\tau}\ln\big(\mathcal{R}^2 M\big) \;=\; -\frac{d}{d\tau}\ln\Big(\mathcal{R}^2\sqrt{m^2+2m\mathcal{Q}}\Big). \tag{18}$$

The expansion is fixed along the flow by the amplitude and the quantum potential, with no free constant, and

$$\frac{d\theta}{d\tau} = -\frac{d^2}{d\tau^2}\ln\big(\mathcal{R}^2 M\big), \tag{19}$$

is the once-differentiated continuity relation. Any statement of the form $d\theta/d\tau = \mathcal{F}$ along this congruence is a rewriting of (19); in particular the Raychaudhuri equation (30) below, a geometric identity for any congruence [13, 16], imposes nothing on θ once (18) holds, and the QRE transcribed to the normalized flow carries no information about θ beyond charge conservation.

3.2 The unnormalized expansion

The expansion $\theta_{\rm D} \equiv \Box S/m$ of the unnormalized flow of Ref. [6] is fixed in the same way. Expanding (9),

$$\mathcal{R}^2\Box S + \nabla_\mu\big(\mathcal{R}^2\big)\nabla^\mu S = 0, \tag{20}$$

and dividing by $\mathcal{R}^2$,

$$\Box S = -\frac{\nabla_\mu(\mathcal{R}^2)}{\mathcal{R}^2}\,\nabla^\mu S = -\nabla_\mu\big(\ln\mathcal{R}^2\big)\,M\hat{u}^\mu, \tag{21}$$

using $\nabla^\mu S = M\hat{u}^\mu$. With $\hat{u}^\mu\nabla_\mu = d/d\tau$,

$$\theta_{\rm D} \;=\; \frac{\Box S}{m} \;=\; -\,\frac{M}{m}\,\frac{d}{d\tau}\ln\mathcal{R}^2. \tag{22}$$

This follows from continuity alone and holds off shell with respect to (10). Both candidate expansions are fixed by the current, and neither carries dynamics of its own.

3.3 Vorticity, acceleration and the Raychaudhuri equation

Three further identities follow from the gradient character of $\hat{u}_\mu$, pointwise on $\mathcal{D}$ in an arbitrary curved spacetime and without use of the Einstein equation. Differentiating (12),

$$\nabla_\nu\hat{u}_\mu = \frac{\nabla_\nu\nabla_\mu S}{M} - \frac{\nabla_\mu S\,\nabla_\nu M}{M^2} = \frac{\nabla_\nu\nabla_\mu S}{M} - \hat{u}_\mu\nabla_\nu\ln M. \tag{23}$$

Since $\nabla_{[\nu}\nabla_{\mu]}S = 0$ for a scalar and a torsion-free connection, the first term drops out of the antisymmetric part,

$$\nabla_{[\nu}\hat{u}_{\mu]} = -\hat{u}_{[\mu}\nabla_{\nu]}\ln M. \tag{24}$$

which $h_\mu{}^\alpha h_\nu{}^\beta$ annihilates because $h_\mu{}^\alpha \hat{u}_\alpha = 0$. Hence

$$\omega_{\mu\nu} \equiv h_\mu{}^\alpha h_\nu{}^\beta \nabla_{[\beta}\hat{u}_{\alpha]} = 0. \tag{25}$$

as Frobenius' theorem requires for a normalized gradient. (Reference [12] reports a nonvanishing rotation tensor here; the gradient structure (12) excludes it.) Contracting (23) with $\hat{u}^\nu$,

$$a_\mu \equiv \hat{u}^\nu \nabla_\nu \hat{u}_\mu = \frac{\hat{u}^\nu \nabla_\nu \nabla_\mu S}{M} - \hat{u}_\mu \frac{d}{d\tau} \ln M. \tag{26}$$

and the numerator of the first term is

$$\begin{aligned} \hat{u}^\nu \nabla_\nu \nabla_\mu S &= \hat{u}^\nu \nabla_\mu \nabla_\nu S \\ &= \nabla_\mu \big(\hat{u}^\nu \nabla_\nu S\big) - \big(\nabla_\mu \hat{u}^\nu\big) \nabla_\nu S \\ &= \nabla_\mu \big(M\, \hat{u}^\nu \hat{u}_\nu\big) - M\, \hat{u}_\nu \nabla_\mu \hat{u}^\nu \\ &= -\nabla_\mu M - \tfrac{1}{2} M\, \nabla_\mu \big(\hat{u}^\nu \hat{u}_\nu\big) \\ &= -\nabla_\mu M. \end{aligned} \tag{27}$$

using the symmetry of $\nabla_\nu \nabla_\mu S$, the Leibniz rule, $\nabla_\nu S = M \hat{u}_\nu$, and the constancy of $\hat{u}^\nu \hat{u}_\nu = -1$. Substituting (27) into (26),

$$\begin{aligned} a_\mu &= -\frac{\nabla_\mu M}{M} - \hat{u}_\mu \frac{d}{d\tau} \ln M \\ &= -\nabla_\mu \ln M - \hat{u}_\mu \hat{u}^\nu \nabla_\nu \ln M \\ &= -\big(\delta_\mu{}^\nu + \hat{u}_\mu \hat{u}^\nu\big) \nabla_\nu \ln M \\ &= -h_\mu{}^\nu \nabla_\nu \ln M, \end{aligned} \tag{28}$$

so that $a_\mu \hat{u}^\mu = 0$: the acceleration is purely spatial. With $M^2 = m^2 + 2m\mathcal{Q}$,

$$\begin{aligned} a_\mu &= -\tfrac{1}{2}\, h_\mu{}^\nu \nabla_\nu \ln\big(m^2 + 2m\mathcal{Q}\big) \\ &\simeq -\frac{1}{m}\, h_\mu{}^\nu \nabla_\nu \mathcal{Q} \qquad (|\mathcal{Q}| \ll m), \end{aligned} \tag{29}$$

the relativistic quantum force, obtained here from the normalization of the flow rather than postulated. The Raychaudhuri equation for a normalized, accelerated congruence [2, 13] then reads

$$\frac{d\theta}{d\tau} = -\frac{\theta^2}{3} - \sigma_{\mu\nu}\sigma^{\mu\nu} - R_{\mu\nu}\hat{u}^\mu \hat{u}^\nu + \nabla_\mu a^\mu, \tag{30}$$

with the vorticity term absent by (25). Taking the divergence of $a^\mu = -\nabla^\mu \ln M - \hat{u}^\mu\, d\ln M/d\tau$ term by term,

$$\begin{aligned} \nabla_\mu a^\mu = &- \Box \ln M - \big(\nabla_\mu \hat{u}^\mu\big) \frac{d}{d\tau} \ln M \\ &- \hat{u}^\mu \nabla_\mu \Big(\frac{d}{d\tau} \ln M\Big), \end{aligned} \tag{31}$$

that is,

$$\nabla_\mu a^\mu = -\Box \ln M - \theta\, \frac{d}{d\tau} \ln M - \frac{d^2}{d\tau^2} \ln M. \tag{32}$$

These identities were verified by exact symbolic computation for a generic four-dimensional phase function on a generic curved metric.

Equation (25) bears on the no-crossing argument of Ref. [6]. For a congruence, no-crossing is hypersurface orthogonality, whose consequence is precisely $\omega_{\mu\nu} = 0$. On the right of (30), $-\theta^2/3$ and $-\sigma_{\mu\nu}\sigma^{\mu\nu}$ are non-positive, $-R_{\mu\nu}\hat{u}^\mu\hat{u}^\nu$ has a sign fixed by the matter content, and $\nabla_\mu a^\mu$ has none; only $+\omega_{\mu\nu}\omega^{\mu\nu}$, absent here, is definitely defocusing. Whatever prevents or permits focusing must therefore come from $\nabla_\mu a^\mu$ or $R_{\mu\nu}\hat{u}^\mu\hat{u}^\nu$.

3.4 What the identity does and does not determine

Equation (18) instantiates a familiar relation: continuity gives $\theta = -d\ln n/d\tau$ for any timelike congruence transporting a conserved density, dust and a baryonic perfect fluid included [15]. What matters here is what it says about the accelerated terms of the Raychaudhuri equation. The congruence and the current share one generating function,

$$\hat{u}^\mu = \frac{\nabla^\mu S}{M}, \qquad j^\mu = \frac{\mathcal{R}^2 M}{\hbar}\,\hat{u}^\mu, \qquad M = \sqrt{-(\nabla S)^2}, \tag{33}$$

so the congruence has no existence apart from the current, and its expansion follows from (18) once the field is known. At $\mathcal{Q} = 0$ the effective mass is constant, the acceleration vanishes, and (18) reduces to the statement for irrotational dust. For $\mathcal{Q} \neq 0$ the flow is accelerated, and the terms by which the QRE differs from the geodesic equation are collected in $\nabla_\mu a^\mu$, Eq. (32), which has no definite sign. For a geodesic, irrotational congruence obeying the strong energy condition the Raychaudhuri identity closes into $d\theta/d\tau \leq -\theta^2/3$ and yields focusing without solving the matter equations; this is what allows the classical equation to support singularity theorems. Here it does not close, and (18) states what the indefinite terms amount to in aggregate.

Conclusions about focusing, conjugate points and singularity avoidance are therefore statements about the growth of $\mathcal{R}^2 M$, to be read off the QHJ equation (10) and the Einstein equation rather than off the QRE. This makes exact the observation of Ref. [23] that the cosmological application rests on a kinematical identity, and it is consistent with Ref. [24]: whatever the quantum corrections, the expansion is fixed by the current.

3.5 Where unbounded defocusing can occur

For a smooth solution $n = \mathcal{R}^2 M/\hbar$ is smooth and positive on $\mathcal{D}$ away from nodes, so $\theta = -\dot{n}/n$ can diverge to $+\infty$ at finite proper time only if $n \to 0$, that is where

$$\mathcal{R} \to 0 \quad \text{(a node)} \qquad \text{or} \qquad M \to 0 \quad (\mathcal{Q} \to -m/2), \tag{34}$$

the second alternative by (11). These are the two loci at which the Madelung–Bohm description degenerates: at a node $\nabla_\mu S$ is singular in the hydrodynamic variables, and at $M = 0$ the flow leaves the timelike cone and $\mathcal{D}$ ends [10, 12]. Unbounded defocusing is thus the divergence of the congruence's own normalization at the boundary of its validity. Caustic behavior, $\theta \to -\infty$, corresponds to $d\ln n/d\tau \to +\infty$, the ordinary growth of density under focusing, which (18) does not forbid.

4 The Field as a Gravitating Source

The identities of Sec. 3 hold on any background, but trajectories on a fixed geometry cannot alter the curvature that defines a singularity [12]. We therefore let the field source the geometry, $G_{\mu\nu} = 8\pi G\, T_{\mu\nu}$.

4.1 Stress tensor in polar variables

Varying (5) with respect to the inverse metric,

$$T_{\mu\nu} = 2\nabla_{(\mu}\phi^*\nabla_{\nu)}\phi - g_{\mu\nu}\Big[\nabla_\alpha\phi^*\nabla^\alpha\phi + \mu^2\phi^*\phi\Big]. \tag{35}$$

With (6) the phase factors cancel in every bilinear,

$$\begin{aligned}\nabla_\mu\phi^*\nabla_\nu\phi &= \Big(\nabla_\mu\mathcal{R} - \frac{i}{\hbar}\mathcal{R}\nabla_\mu S\Big)\Big(\nabla_\nu\mathcal{R} + \frac{i}{\hbar}\mathcal{R}\nabla_\nu S\Big)\\ &= \nabla_\mu\mathcal{R}\nabla_\nu\mathcal{R} + \frac{\mathcal{R}^2}{\hbar^2}\nabla_\mu S\nabla_\nu S\\ &+ \frac{i\mathcal{R}}{\hbar}\big(\nabla_\mu\mathcal{R}\nabla_\nu S - \nabla_\nu\mathcal{R}\nabla_\mu S\big),\end{aligned} \tag{36}$$

the imaginary part being removed by the symmetrization in (35). Its trace gives $\nabla_\alpha\phi^*\nabla^\alpha\phi = (\nabla\mathcal{R})^2 + \mathcal{R}^2(\nabla S)^2/\hbar^2$, and $\phi^*\phi = \mathcal{R}^2$, so

$$\begin{aligned}T_{\mu\nu} &= 2\nabla_\mu\mathcal{R}\nabla_\nu\mathcal{R} + \frac{2}{\hbar^2}\mathcal{R}^2\nabla_\mu S\nabla_\nu S\\ &- g_{\mu\nu}\Big[(\nabla\mathcal{R})^2 + \frac{\mathcal{R}^2(\nabla S)^2}{\hbar^2} + \mu^2\mathcal{R}^2\Big],\end{aligned} \tag{37}$$

exact and off shell. On shell, with $\nabla_\mu S = M\hat{u}_\mu$ and (10)–(11), the bracket is

$$\begin{aligned}(\nabla\mathcal{R})^2 + \frac{\mathcal{R}^2(\nabla S)^2}{\hbar^2} + \mu^2\mathcal{R}^2 &= (\nabla\mathcal{R})^2 - \frac{\mathcal{R}^2M^2}{\hbar^2} + \frac{m^2\mathcal{R}^2}{\hbar^2}\\ &= (\nabla\mathcal{R})^2 - \frac{2m\mathcal{Q}\mathcal{R}^2}{\hbar^2},\end{aligned} \tag{38}$$

so that

$$\begin{aligned}T_{\mu\nu} &= 2\nabla_\mu\mathcal{R}\nabla_\nu\mathcal{R} + \frac{2\mathcal{R}^2M^2}{\hbar^2}\hat{u}_\mu\hat{u}_\nu\\ &- g_{\mu\nu}\Big[(\nabla\mathcal{R})^2 - \frac{2m\mathcal{Q}\mathcal{R}^2}{\hbar^2}\Big].\end{aligned} \tag{39}$$

The quantum potential enters in two places: through M^2 in the term along $\hat{u}_\mu\hat{u}_\nu$, and explicitly in the bracket multiplying $g_{\mu\nu}$. Two contractions of (37) control what follows.

4.2 The null channel

Contracting (37) with a null k^μ term by term,

$$\begin{aligned}T_{\mu\nu}k^\mu k^\nu &= 2\big(k^\mu\nabla_\mu\mathcal{R}\big)\big(k^\nu\nabla_\nu\mathcal{R}\big)\\ &+ \frac{2}{\hbar^2}\mathcal{R}^2\big(k^\mu\nabla_\mu S\big)\big(k^\nu\nabla_\nu S\big)\\ &- \big(g_{\mu\nu}k^\mu k^\nu\big)\Big[(\nabla\mathcal{R})^2 + \frac{\mathcal{R}^2(\nabla S)^2}{\hbar^2} + \mu^2\mathcal{R}^2\Big]\\ &= 2\big(k^\mu\nabla_\mu\mathcal{R}\big)^2 + \frac{2}{\hbar^2}\mathcal{R}^2\big(k^\mu\nabla_\mu S\big)^2,\end{aligned} \tag{40}$$

the third line vanishing because $g_{\mu\nu}k^\mu k^\nu = 0$. Both surviving terms are squares with non-negative coefficients, so

$$T_{\mu\nu}k^\mu k^\nu = 2(k^\mu\nabla_\mu\mathcal{R})^2 + \frac{2}{\hbar^2}\mathcal{R}^2(k^\mu\nabla_\mu S)^2 \;\geq\; 0, \tag{41}$$

off shell and for every state.

That a minimally coupled scalar obeys the null energy condition is standard [25]; what matters is how the quantum potential enters. The null contraction removes the bracket of (39), and with it the explicit quantum-potential term, while on shell the second square in (41) is

$$\frac{2}{\hbar^2}\mathcal{R}^2(k^\mu\nabla_\mu S)^2 = \frac{2\mathcal{R}^2M^2}{\hbar^2}\,(k^\mu\hat{u}_\mu)^2. \tag{42}$$

It survives in the null channel only through the positive factor M^2 multiplying a square, so it can change the strength of null focusing but never its sign, for any state, and cannot produce the null defocusing needed to evade the Penrose theorem [3].

4.3 The timelike channel

The trace of (37), with $g^{\mu\nu}g_{\mu\nu} = 4$, is

$$\begin{aligned} T &= 2(\nabla\mathcal{R})^2 + \frac{2}{\hbar^2}\mathcal{R}^2(\nabla S)^2 \\ &\quad - 4\Big[(\nabla\mathcal{R})^2 + \frac{\mathcal{R}^2(\nabla S)^2}{\hbar^2} + \mu^2\mathcal{R}^2\Big] \\ &= -2(\nabla\mathcal{R})^2 - \frac{2}{\hbar^2}\mathcal{R}^2(\nabla S)^2 - 4\mu^2\mathcal{R}^2. \end{aligned} \tag{43}$$

and, with $\dot{\mathcal{R}} \equiv \hat{u}^\mu\nabla_\mu\mathcal{R}$, $\hat{u}^\mu\nabla_\mu S = -M$ and $\hat{u}^\mu\hat{u}_\mu = -1$,

$$\begin{aligned} T_{\mu\nu}\hat{u}^\mu\hat{u}^\nu &= 2\dot{\mathcal{R}}^2 + \frac{2}{\hbar^2}\mathcal{R}^2M^2 \\ &\quad + \;(\nabla\mathcal{R})^2 + \frac{\mathcal{R}^2(\nabla S)^2}{\hbar^2} + \mu^2\mathcal{R}^2. \end{aligned} \tag{44}$$

Since $R_{\mu\nu}\hat{u}^\mu\hat{u}^\nu = 8\pi G\big(T_{\mu\nu} - \tfrac{1}{2}g_{\mu\nu}T\big)\hat{u}^\mu\hat{u}^\nu$ and $g_{\mu\nu}\hat{u}^\mu\hat{u}^\nu = -1$, the second piece is $+\tfrac{1}{2}T$, and adding (44) to half of (43),

$$\begin{aligned} &\big(T_{\mu\nu} - \tfrac{1}{2}g_{\mu\nu}T\big)\hat{u}^\mu\hat{u}^\nu \\ &\quad= \underbrace{2\dot{\mathcal{R}}^2 + \frac{2}{\hbar^2}\mathcal{R}^2M^2 + (\nabla\mathcal{R})^2 + \frac{\mathcal{R}^2(\nabla S)^2}{\hbar^2} + \mu^2\mathcal{R}^2}_{\text{Eq. (44)}} \\ &\qquad+ \underbrace{\Big[-(\nabla\mathcal{R})^2 - \frac{\mathcal{R}^2(\nabla S)^2}{\hbar^2} - 2\mu^2\mathcal{R}^2\Big]}_{\frac{1}{2}\times\text{Eq. (43)}} \\ &\quad= 2\dot{\mathcal{R}}^2 + \frac{2}{\hbar^2}\mathcal{R}^2M^2 - \mu^2\mathcal{R}^2, \end{aligned} \tag{45}$$

the gradient terms cancelling pairwise. The on-shell value of $(\nabla S)^2$ is never needed, and the only surviving dependence on the quantum potential is the M^2 carried by $(\hat{u}^\mu\nabla_\mu S)^2$. Multiplying by

$8\pi G$ and using $\mu^2 = m^2/\hbar^2$, on $\mathcal{D}$,

$$R_{\mu\nu}\hat{u}^\mu\hat{u}^\nu = 8\pi G\left[2\dot{\mathcal{R}}^2 + \frac{\mathcal{R}^2}{\hbar^2}\left(2M^2 - m^2\right)\right], \tag{46}$$

in which the quantum potential enters only through the non-negative term $2\mathcal{R}^2M^2/\hbar^2$.

Two consequences follow. Dropping $2\mathcal{R}^2M^2/\hbar^2 > 0$ from (46), and then $2\dot{\mathcal{R}}^2 \geq 0$,

$$R_{\mu\nu}\hat{u}^\mu\hat{u}^\nu \;\geq\; 8\pi G\left[2\dot{\mathcal{R}}^2 - \mu^2\mathcal{R}^2\right] \;\geq\; -8\pi G\,\mu^2\mathcal{R}^2, \tag{47}$$

where the first inequality is approached only as $M \to 0$, that is on $\partial\mathcal{D}$, and the second only where in addition $\dot{\mathcal{R}} = 0$.

Second, timelike defocusing, $R_{\mu\nu}\hat{u}^\mu\hat{u}^\nu < 0$, requires, on dividing (46) by $8\pi G\mathcal{R}^2/\hbar^2 > 0$,

$$2M^2 - m^2 < -\frac{2\hbar^2\dot{\mathcal{R}}^2}{\mathcal{R}^2}, \tag{48}$$

that is $M^2 < m^2/2 - \hbar^2\dot{\mathcal{R}}^2/\mathcal{R}^2$, so that with $M^2 = m^2 + 2m\mathcal{Q}$,

$$2m\mathcal{Q} < -\frac{m^2}{2} - \frac{\hbar^2\dot{\mathcal{R}}^2}{\mathcal{R}^2} \iff \mathcal{Q} < -\frac{m}{4} - \frac{\hbar^2}{2m}\left(\frac{\dot{\mathcal{R}}}{\mathcal{R}}\right)^2. \tag{49}$$

With the domain condition (13), defocusing occurs precisely in the window

$$-\frac{m}{2} \;<\; \mathcal{Q} \;<\; -\frac{m}{4} - \frac{\hbar^2}{2m}\left(\frac{\dot{\mathcal{R}}}{\mathcal{R}}\right)^2. \tag{50}$$

which is nonempty iff the lower limit lies below the upper one,

$$-\frac{m}{2} < -\frac{m}{4} - \frac{\hbar^2}{2m}\left(\frac{\dot{\mathcal{R}}}{\mathcal{R}}\right)^2 \iff \left(\frac{\dot{\mathcal{R}}}{\mathcal{R}}\right)^2 < \frac{m^2}{2\hbar^2}, \tag{51}$$

that is, only where

$$\left|\frac{\dot{\mathcal{R}}}{\mathcal{R}}\right| < \frac{m}{\sqrt{2}\,\hbar}. \tag{52}$$

4.4 Physical content of the two channels

The floor in (47), $-8\pi G\mu^2\mathcal{R}^2$, is minus the classical mass-potential term: the strong-energy-condition violation of an ordinary scalar in its potential-dominated regime, the mechanism behind inflation and quintessence [26, 25]. The quantum potential can reduce timelike focusing below its mass-shell value, and in the window (50) it does, but it cannot take the convergence below that classical floor, and approaches it only as $M \to 0$, where the congruence ceases to exist. For $\mathcal{Q} > 0$, hence $M > m$, the same term makes the convergence larger than its classical value without bound. Condition (52) restricts defocusing to amplitudes varying more slowly than the Compton rate $m/\hbar$, up to a factor $\sqrt{2}$: a steeper variation contributes $2\dot{\mathcal{R}}^2$ with a positive sign and closes the window whatever the value of $\mathcal{Q}$, the mechanism by which it shuts near a cosmological singularity (Sec. 5.4).

In both channels, then, the quantum potential enters only through $M^2 > 0$ multiplying a non-negative quantity, and the sole negative contribution is the classical mass term $-\mu^2\mathcal{R}^2$, which survives trace reversal in the timelike channel and is annihilated in the null one.

5 Cosmological Implications

5.1 The congruence in flat FRW

In the cosmological application [11] the normalization issues disappear. Take spatially flat FRW, $ds^2 = -d\tau^2 + a^2(\tau)\, d\vec{x}^2$, with homogeneous $\mathcal{R}(\tau)$ and $S(\tau)$, so that $\nabla_\mu S = (\dot{S}, 0, 0, 0)$ and

$$(\nabla S)^2 = g^{00}\dot{S}^2 = -\dot{S}^2 \;\leq\; 0, \tag{53}$$

The phase gradient is timelike wherever $\dot{S} \neq 0$, with $M = |\dot{S}|$, so homogeneity enforces the domain condition. On the positive-frequency branch $\dot{S} < 0$ and $\hat{u}^\mu = (1, \vec{0})$: the normalized congruence is the comoving frame,

$$\begin{aligned} \hat{u}^\mu &= (1,0,0,0), \qquad & a_\mu &= 0, \\ \sigma_{\mu\nu} &= \omega_{\mu\nu} = 0, \qquad & \theta &= 3H. \end{aligned} \tag{54}$$

with $a_\mu = 0$ by (28) because M is spatially constant, and $\theta = a^{-3}\partial_\tau(a^3) = 3H$. The congruence is unit, timelike, hypersurface orthogonal and geodesic, and the Bohmian and classical congruences coincide.

5.2 Source terms and the conserved charge

With $(\nabla\mathcal{R})^2 = -\dot{\mathcal{R}}^2$ and $(\nabla S)^2 = -\dot{S}^2$, (37) gives $\rho = T_{00}$,

$$\begin{aligned} \rho &= 2\dot{\mathcal{R}}^2 + \frac{2}{\hbar^2}\mathcal{R}^2\dot{S}^2 + \Big[-\dot{\mathcal{R}}^2 - \frac{\mathcal{R}^2\dot{S}^2}{\hbar^2} + \mu^2\mathcal{R}^2 \Big] \\ &= \dot{\mathcal{R}}^2 + \frac{\mathcal{R}^2\dot{S}^2}{\hbar^2} + \mu^2\mathcal{R}^2. \end{aligned} \tag{55}$$

and, since homogeneity kills the first two terms of (37) in the spatial components,

$$\begin{aligned} T_{ij} &= -g_{ij}\Big[(\nabla\mathcal{R})^2 + \frac{\mathcal{R}^2(\nabla S)^2}{\hbar^2} + \mu^2\mathcal{R}^2\Big] \\ &= -g_{ij}\Big[-\dot{\mathcal{R}}^2 - \frac{\mathcal{R}^2\dot{S}^2}{\hbar^2} + \mu^2\mathcal{R}^2\Big] \\ &= g_{ij}\Big[\dot{\mathcal{R}}^2 + \frac{\mathcal{R}^2\dot{S}^2}{\hbar^2} - \mu^2\mathcal{R}^2\Big]. \end{aligned} \tag{56}$$

so that $T_{ij} = p\, g_{ij}$ with

$$p = \dot{\mathcal{R}}^2 + \frac{\mathcal{R}^2\dot{S}^2}{\hbar^2} - \mu^2\mathcal{R}^2. \tag{57}$$

Hence

$$\rho + p = 2\dot{\mathcal{R}}^2 + \frac{2\mathcal{R}^2\dot{S}^2}{\hbar^2}, \qquad \rho - p = 2\mu^2\mathcal{R}^2. \tag{58}$$

Combining (55) and (57), $\rho + 3p = 4\dot{\mathcal{R}}^2 + 4\mathcal{R}^2\dot{S}^2/\hbar^2 - 2\mu^2\mathcal{R}^2$, so

$$4\pi G(\rho + 3p) = 8\pi G\left[2\dot{\mathcal{R}}^2 + \frac{\mathcal{R}^2}{\hbar^2}\big(2M^2 - m^2\big)\right], \tag{59}$$

reproducing (46) with $M = |\dot{S}|$, as it must. The continuity equation reduces to $a^{-3}\partial_\tau(a^3\mathcal{R}^2\dot{S}) = 0$ and integrates to

$$\mathcal{R}^2 M a^3 \;=\; C \;=\; \text{const} \;\geq\; 0, \tag{60}$$

the conserved $U(1)$ charge per comoving volume; these homogeneous equations are those of charged scalar-field cosmology in hydrodynamic form [20, 21]. The central identity now reads

$$3H = -\frac{d}{d\tau}\ln\!\big(\mathcal{R}^2 M\big) = -\frac{d}{d\tau}\ln\!\Big(\frac{C}{a^3}\Big) = 3\,\frac{d}{d\tau}\ln a, \tag{61}$$

an identity: it is (60) differentiated. A quantum-corrected Friedmann equation obtained by substituting the QRE for the second Friedmann equation [11] is therefore not an independent dynamical equation for the minimally coupled field, whose dynamics resides in the QHJ equation and the Friedmann constraint,

$$H^2 = \frac{8\pi G}{3}\rho, \tag{62}$$

with ρ from (55). Equation (60) also fixes

$$M = \frac{C}{\mathcal{R}^2 a^3}, \qquad \mathcal{Q} = \frac{M^2 - m^2}{2m} = \frac{1}{2m}\Big(\frac{C^2}{\mathcal{R}^4 a^6} - m^2\Big), \tag{63}$$

so the quantum potential is not a free function available for tuning against the singularity: it is set by the amplitude and the scale factor.

5.3 The dust-like floor and the absence of a bounce

The conserved charge divides the minimally coupled system into two cases.

If $C = 0$, then with $a > 0$ we have $\dot{S} = 0$ wherever $\mathcal{R} \neq 0$, hence $M = 0$: the current vanishes and the Bohmian congruence (12) does not exist, so the QRE has no congruence to describe.

If $C \neq 0$, three properties follow. First, with $\dot{H} = -4\pi G(\rho + p)$ and (58),

$$\dot{H} = -4\pi G\Big(2\dot{\mathcal{R}}^2 + \frac{2\mathcal{R}^2\dot{S}^2}{\hbar^2}\Big) \;<\; 0, \tag{64}$$

strictly, because $C \neq 0$ implies $\mathcal{R}^2\dot{S}^2 > 0$ everywhere by (60). Second, the arithmetic-geometric mean inequality applied to the two terms of (55) other than $\dot{\mathcal{R}}^2$ gives

$$\frac{\mathcal{R}^2\dot{S}^2}{\hbar^2} + \mu^2\mathcal{R}^2 \;\geq\; 2\sqrt{\frac{\mathcal{R}^2\dot{S}^2}{\hbar^2}\,\mu^2\mathcal{R}^2} = \frac{2m\mathcal{R}^2|\dot{S}|}{\hbar^2}, \tag{65}$$

with equality iff $M = |\dot{S}| = m$, that is $\mathcal{Q} = 0$ by (11). Equivalently the difference is a perfect square,

$$\frac{\mathcal{R}^2\dot{S}^2}{\hbar^2} + \mu^2\mathcal{R}^2 - \frac{2m\mathcal{R}^2 M}{\hbar^2} = \frac{\mathcal{R}^2}{\hbar^2}\big(M - m\big)^2 \;\geq\; 0. \tag{66}$$

and adding $\dot{\mathcal{R}}^2 \geq 0$ and using (60) gives the dust-like floor

$$\rho \;\geq\; \frac{2m}{\hbar^2}\,\mathcal{R}^2 M \;=\; \frac{2mC}{\hbar^2 a^3}, \tag{67}$$

with equality only where $\mathcal{Q} = 0$ and $\dot{\mathcal{R}} = 0$. Since the rest-frame particle number density is $2n/\hbar = 2\mathcal{R}^2 M/\hbar^2$, this states that ρ is at least the rest-mass density of the conserved charge, the energy density of pressureless dust. With the Friedmann constraint (62),

$$H^2 = \frac{8\pi G}{3}\rho \;\geq\; \frac{8\pi G}{3}\cdot\frac{2mC}{\hbar^2 a^3} = \frac{16\pi G\, mC}{3\hbar^2 a^3} \;>\; 0, \tag{68}$$

so H never vanishes. A bounce requires H to pass from negative to positive and a recollapse from positive to negative; both require $H = 0$, which (68) excludes. With (64), H is strictly monotonic and of fixed sign: the universe is either always expanding or always contracting. Third, on the contracting branch (68) gives $\dot{a} = aH \leq -\kappa\, a^{-1/2}$ with $\kappa \equiv \sqrt{16\pi G m C/(3\hbar^2)}$, so

$$\frac{d}{d\tau}\big(a^{3/2}\big) = \frac{3}{2}a^{1/2}\dot{a} \;\leq\; -\frac{3}{2}\kappa, \tag{69}$$

and $a^{3/2}$ reaches zero from $a_0^{3/2}$ within

$$\Delta\tau \;\leq\; \frac{2}{3}\sqrt{\frac{3\hbar^2}{16\pi G\, mC}}\; a_0^{3/2}. \tag{70}$$

As $a \to 0$, (67) and (68) force $\rho \to \infty$ and $\theta = 3H \to -\infty$; the expanding branch traced backward emerges from $a = 0$ within the same bound. Since the congruence is comoving, τ is proper time along the Bohmian trajectories themselves.

The mechanism is the one encoded in (67): the conserved quanta, compressed into a shrinking comoving volume, gravitate at least as pressureless dust, and the quantum potential does not resist. By (63), for bounded amplitude M grows without bound as $a \to 0$ and then strengthens focusing through (46). The flow lines do not cross; they converge simultaneously onto $a = 0$, a homogeneous caustic, in finite proper time.

5.4 The defocusing window near $a = 0$

Sustained defocusing would require an amplitude growing fast enough to keep $\mathcal{Q}$ negative. By (63), $\mathcal{Q} < 0$ requires $M < m$, that is

$$\mathcal{R}^2 a^3 > \frac{C}{m}, \tag{71}$$

so as $a \to 0$ the amplitude must grow faster than $a^{-3/2}$ and its logarithmic growth rate is, on average, at least of order $\frac{3}{2}|H|$. Since (52) requires $|\dot{\mathcal{R}}/\mathcal{R}| < m/(\sqrt{2}\hbar)$, an open window becomes incompatible with sustained $\mathcal{Q} < 0$ once

$$\frac{3}{2}|H| \;\gtrsim\; \frac{m}{\sqrt{2}\,\hbar} \qquad\Longleftrightarrow\qquad |H| \;\gtrsim\; \frac{\sqrt{2}\,m}{3\hbar}. \tag{72}$$

which by the floor (68) is guaranteed once

$$\frac{16\pi G\, mC}{3\hbar^2 a^3} \;\gtrsim\; \frac{2m^2}{9\hbar^2} \qquad\Longleftrightarrow\qquad a \;\lesssim\; \left(\frac{24\pi G\, C}{m}\right)^{1/3}. \tag{73}$$

Because it compares an average growth rate with a pointwise condition, this estimate locates the scale at which the window closes on the approach to $a = 0$ rather than providing a sharp bound.

5.5 Spatial curvature, additional matter, and non-minimal coupling

For $k = +1$ a bounce requires $\frac{8\pi G}{3}\rho = a^{-2}$, which a classical potential-dominated scalar can arrange [27]; with the floor (67),

$$\frac{16\pi G\, mC}{3\hbar^2 a^3} \;\leq\; \frac{1}{a^2} \qquad\Longleftrightarrow\qquad a \;\geq\; \frac{16\pi G\, mC}{3\hbar^2}, \tag{74}$$

so small-radius bounces are excluded, and any large-radius bounce is the classical scalar mechanism, owing nothing to $\mathcal{Q}$. Additional matter satisfying the null energy condition only strengthens (64)

and the finite-time conclusion. The cosmological-constant-like behavior reported in Ref. [11] survives here only as the $w \to -1$ potential-dominated regime of the classical scalar [23]; by (58) nothing in it is specific to the quantum potential.

The infinite-age result of Ref. [11] is driven not by the quantum potential but by the ξR term, which on FRW contributes at order H^4. That term is absent at $\xi = 0$, so the present results neither reproduce nor refute it; they establish the complementary statement that the quantum potential of the minimally coupled field cannot avert the singularity.

6 Discussion and Conclusion

6.1 Relation to other approaches

The normalized congruence and its Raychaudhuri equation were written down in Ref. [12]. The present analysis identifies that congruence as the unit flow of the conserved current, derives the closed form (18) of its expansion, and carries out the focusing analysis that this makes possible, with full back-reaction.

Several related programmes lie outside these conclusions. The quantized geodesic congruence of Ref. [16] is a different object from the classical hydrodynamic congruence of a Bohmian matter field, and the two analyses are logically independent. The fixed-background study of the Bohmian deviation equation [17] reached a negative verdict on singularity avoidance by the trajectory route; the present results reach the same verdict with back-reaction included and locate its mechanism in the conserved current. Quantized metric fluctuations strengthen focusing in both the timelike and the null Raychaudhuri equations [18, 19]; combined with (41) and (47), quantized geometry assists focusing while minimally coupled Bohmian matter cannot oppose it beyond the classical potential budget and cannot reverse null focusing at all.

The sharpest contrast is with Bohmian quantum cosmology [28], where the Wheeler–DeWitt wave function of minisuperspace is given a de Broglie–Bohm reading, the quantum potential acts on the scale factor itself, and bounces are well documented. Section 5 shows that no such rescue is available when the Bohmian field is minimally coupled matter on a classical geometry: such a mechanism must act on the geometry, not through the matter sector.

6.2 Limitations

Four limitations bound these claims. First, the field is a classical complex scalar, the object underlying Refs. [6, 11], not a second-quantized field; renormalized stress tensors can violate the null energy condition, and the appropriate statements there are the quantum energy inequalities and the quantum focusing conjecture [25, 29], which lie outside this scope. Second, the cosmological results are exact only under FRW symmetry; the identities of Secs. 3 and 4 hold in any spacetime, but no claim is made about generic inhomogeneous collapse beyond the bound (47). Third, (46)–(50) use the comoving frame and hence the domain $\mathcal{D}$, outside which the phase flow defines no timelike congruence in any normalization.

Fourth, the curvature coupling $\xi R|\phi|^2$ has been set to zero, which is not an innocuous simplification: it enters the Klein–Gordon equation as $(\Box - m^2/\hbar^2 + \xi R)\phi = 0$ and on FRW contributes the H^4 correction on which the infinite-age mechanism of Ref. [11] rests. The present results establish that the quantum potential alone, in the minimally coupled theory, cannot avert the singularity; they should not be read as resolving that mechanism, which requires separate analysis.

6.3 Conclusion

The normalized Bohmian phase flow is the unit flow of the conserved Klein–Gordon current, so its expansion is fixed by charge conservation, $\theta = -d\ln(\mathcal{R}^2 M)/d\tau$; the unnormalized expansion is fixed in the same way. Evaluated on this congruence the quantum Raychaudhuri equation is the once-differentiated continuity relation rather than an independent evolution equation for θ, and since the flow is accelerated wherever $\mathcal{Q} \neq 0$ the Raychaudhuri identity does not close into a focusing inequality. Focusing must therefore be settled by the Klein–Gordon and Einstein equations.

For the minimally coupled complex scalar the picture is consistent. The flow has zero vorticity, so the no-crossing property removes rather than supplies the only term of definite defocusing sign, and unbounded defocusing is confined to wave-function nodes and to the vanishing of the effective mass, where the hydrodynamic description degenerates. With the field gravitating, the quantum potential enters both channels only through the positive factor M^2: it cannot reverse null focusing for any state, and cannot take the timelike convergence below $-8\pi G\mu^2\mathcal{R}^2$, a value approached only where the congruence ceases to exist. In flat FRW cosmology either the conserved charge vanishes and there is no Bohmian congruence, or it does not and charge conservation installs a dust-like floor on the energy density that makes the Hubble rate strictly monotonic, excludes any bounce or recollapse, and drives the scale factor to zero in finite proper time along the congruence itself. The non-minimal ξR mechanism, which acts through an H^4 term rather than through the quantum potential, remains open.

Acknowledgments

The author thanks the Jamal Nazrul Islam Center for Advanced Research, University of Chittagong, for institutional support. This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors. No new data were created or analyzed in this study; all results are analytic and reproducible from the equations given in the text.

References

[1] A. K. Raychaudhuri, *Relativistic cosmology I*, Phys. Rev. **98**, 1123 (1955).

[2] S. Kar and S. SenGupta, *The Raychaudhuri equations: A brief review*, Pramana **69**, 49 (2007), arXiv:gr-qc/0611123.

[3] R. Penrose, *Gravitational collapse and space-time singularities*, Phys. Rev. Lett. **14**, 57 (1965).

[4] S. W. Hawking and R. Penrose, *The singularities of gravitational collapse and cosmology*, Proc. R. Soc. A **314**, 529 (1970).

[5] J. M. M. Senovilla and D. Garfinkle, *The 1965 Penrose singularity theorem*, Class. Quantum Grav. **32**, 124008 (2015), arXiv:1410.5226 [gr-qc].

[6] S. Das, *Quantum Raychaudhuri equation*, Phys. Rev. D **89**, 084068 (2014), arXiv:1311.6539 [gr-qc].

[7] L. de Broglie, *La mécanique ondulatoire et la structure atomique de la matière et du rayonnement*, J. Phys. Radium **8**, 225 (1927).

[8] E. Madelung, *Quantentheorie in hydrodynamischer Form*, Z. Phys. **40**, 322 (1927).

[9] D. Bohm, *A suggested interpretation of the quantum theory in terms of hidden variables I*, Phys. Rev. **85**, 166 (1952).

[10] P. R. Holland, *The Quantum Theory of Motion* (Cambridge University Press, Cambridge, 1993), Chaps. 3 and 12.

[11] A. F. Ali and S. Das, *Cosmology from quantum potential*, Phys. Lett. B **741**, 276 (2015), arXiv:1404.3093 [gr-qc].

[12] E. I. Lashin and D. Dou, *Comment on "Quantum Raychaudhuri equation"*, Phys. Rev. D **95**, 068501 (2017), arXiv:1606.04738 [gr-qc].

[13] R. M. Wald, *General Relativity* (University of Chicago Press, Chicago, 1984), Chap. 9.

[14] S. W. Hawking and G. F. R. Ellis, *The Large Scale Structure of Space-Time* (Cambridge University Press, Cambridge, 1973), Chap. 8.

[15] G. F. R. Ellis, *Republication of: Relativistic cosmology*, Gen. Relativ. Gravit. **41**, 581 (2009).

[16] S. Gupta Choudhury, A. Dasgupta, and N. Banerjee, *The Raychaudhuri equation for a quantized timelike geodesic congruence*, Eur. Phys. J. C **81**, 906 (2021), arXiv:2103.13203 [gr-qc].

[17] F. Rahmani and M. Golshani, *Some clarifications about the Bohmian geodesic deviation equation and Raychaudhuri's equation*, Int. J. Mod. Phys. A **33**, 1850027 (2018), arXiv:1706.08010 [gr-qc].

[18] S.-E. Bak, M. Parikh, S. Sarkar, and F. Setti, *Quantum gravity fluctuations in the timelike Raychaudhuri equation*, JHEP **05** (2023) 125, arXiv:2212.14010 [gr-qc].

[19] S.-E. Bak, M. Parikh, S. Sarkar, and F. Setti, *Quantum-gravitational null Raychaudhuri equation*, JHEP **07** (2024) 214, arXiv:2312.17214 [gr-qc].

[20] B. Li, T. Rindler-Daller, and P. R. Shapiro, *Cosmological constraints on Bose-Einstein-condensed scalar field dark matter*, Phys. Rev. D **89**, 083536 (2014), arXiv:1310.6061 [astro-ph.CO].

[21] A. Suárez and P.-H. Chavanis, *Cosmological evolution of a complex scalar field with repulsive or attractive self-interaction*, Phys. Rev. D **95**, 063515 (2017), arXiv:1608.08624 [gr-qc].

[22] D. H. Delphenich, *The geometrical origin of the Madelung potential*, arXiv:gr-qc/0211065.

[23] E. I. Lashin, *On the correctness of cosmology from quantum potential*, Mod. Phys. Lett. A **31**, 1650044 (2016), arXiv:1505.03070 [gr-qc].

[24] S. Das, *Reply to "Comment on 'Quantum Raychaudhuri equation' "*, Phys. Rev. D **95**, 068502 (2017), arXiv:1702.05219 [gr-qc].

[25] E.-A. Kontou and K. Sanders, *Energy conditions in general relativity and quantum field theory*, Class. Quantum Grav. **37**, 193001 (2020), arXiv:2003.01815 [gr-qc].

[26] C. Barceló and M. Visser, *Twilight for the energy conditions?*, Int. J. Mod. Phys. D **11**, 1553 (2002), arXiv:gr-qc/0205066.

[27] C. Molina-París and M. Visser, *Minimal conditions for the creation of a Friedman–Robertson–Walker universe from a "bounce"*, Phys. Lett. B **455**, 90 (1999), arXiv:gr-qc/9810023.

[28] N. Pinto-Neto and J. C. Fabris, *Quantum cosmology from the de Broglie–Bohm perspective*, Class. Quantum Grav. **30**, 143001 (2013), arXiv:1306.0820 [gr-qc].

[29] R. Bousso, Z. Fisher, S. Leichenauer, and A. C. Wall, *Quantum focusing conjecture*, Phys. Rev. D **93**, 064044 (2016), arXiv:1506.02669 [hep-th].